

Fig. 28.1 One-year survival comparing early goal-directed therapy (EGDT) to usual care in patients with sepsis. Kaplan-Meier survival curves comparing EGDT to usual care in patients with sepsis from pooled patient-level data from the ProCESS, ARISE, and ProMISE trials. Kidney replacement therapy was required in 11.0% of patients who were randomized to EGDT as compared with 10.6% of patients randomized to usual care (OR: 1.02; 95% CI: 0.81–1.28; $P = 0.88$). (From The PRISM Investigators. Goal-directed therapy for septic shock—a patient-level meta-analysis. *N Engl J Med*. 2017;376:2223–2234.)

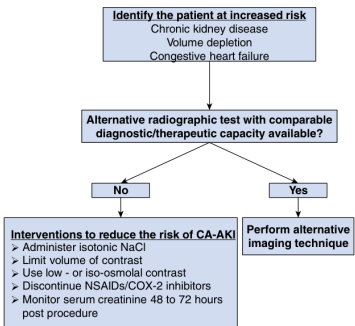


Fig. 28.2 Algorithm for mitigation of risk of contrast-associated acute kidney injury (CA-AKI). (COX-2, Cyclo-oxygenase-2; NSAIDs, nonsteroidal antiinflammatory drugs.)

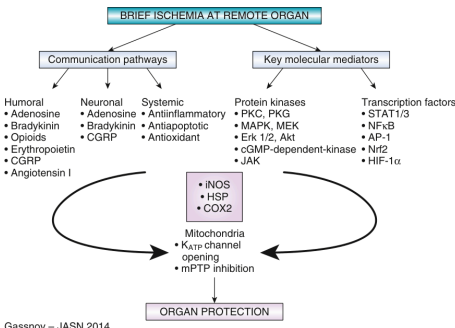


Fig. 28.3 Postulated mechanisms for remote ischemic preconditioning. Postulated mechanisms for renal protective effects of remote ischemic preconditioning. AP-1, Activator protein-1; cGMP, cyclic guanosine monophosphate; CGRP, calcitonin gene-related peptide; COX2, cyclo-oxygenase 2; HIF-1 α , hypoxia-inducible factor 1 α ; HSP, heat shock protein; iNOS, inducible nitric oxide synthase; JAK, Janus kinase; MEK, MAPK kinase; mPTP, mitochondrial permeability transition pore; Nrf2, nuclear factor (erythroid-derived 2)-like 2; STAT1/3, signal transducer and activator of transcription. (From Gassnov N, Na AM, Caglayan E, Et P. Remote ischemic preconditioning and -renoprotection: from myth to a novel therapeutic option? *J Am Soc Nephrol*. 2014;25(2):216–224.)

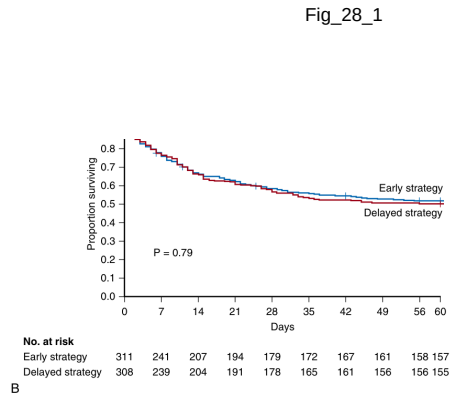


Fig. 28.4 Timing of initiation of kidney replacement therapy (KRT) and survival in the Acute Kidney Initiation in Kidney Injury (AKIKI) trial comparing early and delayed strategies for initiation of KRT. Timing of initiation of KRT (A) and Kaplan-Meier probability of survival (B) in the Acute Kidney Initiation in Kidney Injury (AKIKI) trial. Most (98%) of the patients in the early-treatment group initiated KRT at a median of 4.3 hours after reaching stage 3 AKI as compared with 51% of patients who initiated KRT at a median of 57 hours. Sixty-day mortality was 48.5% in the early-treatment group versus 49.7% in the delayed-treatment group (HR: 1.03; 95% CI: 0.81–1.29; $P = 0.84$). (From Gaudry S, et al. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med*. 2016;375:122–133.)

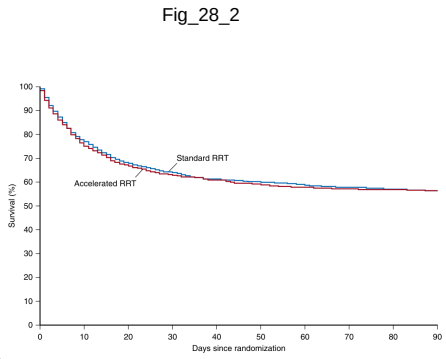


Fig. 28.5 Survival with standard (delayed) versus accelerated initiation of kidney replacement therapy (KRT) in the Standard versus Accelerated Initiation of Renal Replacement Therapy in the Acute Kidney Injury (STARTR-ARI) trial. Kaplan-Meier plot of survival probability among 2827 evaluable patients in the STARTR-ARI trial, which compared strategies of early and delayed KRT in critically ill patients with AKI. Death at 90 days occurred in 653 patients (43.9%) in the accelerated KRT arm as compared with 639 patients (43.7%) in the standard (delayed) KRT arm ($P = 0.92$). RRT, Renal replacement therapy. (From STARTR-ARI Investigators, Australian and New Zealand Intensive Care Society Clinical Trials Group, United Kingdom Critical Care Research Group, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. *N Engl J Med*. 2020;383:240–251.)

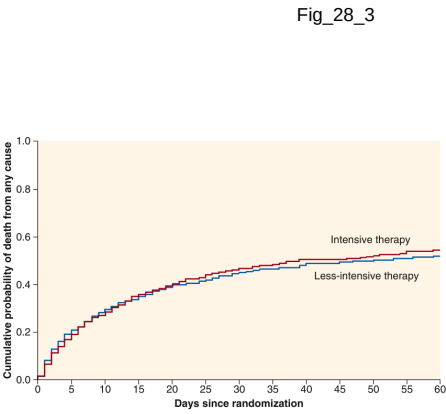


Fig. 28.6 Sixty-day mortality with intensive versus less-intensive kidney replacement therapy in the Acute Renal Failure Trial Network (ATN) study. Kaplan-Meier plot of mortality in 1124 critically ill patients with acute kidney injury randomized to a strategy of more intensive kidney replacement therapy (6x-per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 35 mL/kg per hour) versus less-intensive kidney replacement therapy (3x-per week intermittent hemodialysis or continuous venovenous hemodiafiltration at 20 mL/kg per hour). At 60 days, mortality was 53.6% in the more intensive arm versus 51.5% in the less-intensive arm (OR: 1.09; 95% CI: 0.86–1.40; $P = 0.47$). (From The VA/NIH Acute Renal Failure Trial Network. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med*. 2008; 359:7–20.)

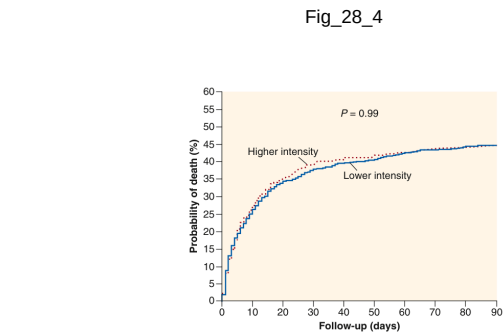


Fig. 28.7 Ninety-day mortality with intensive versus less-intensive continuous venovenous hemodiafiltration (CVVHDF) in the Randomized Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy Study Kaplan-Meier plot of mortality in 1508 critically ill patients with acute kidney injury randomized to CVVHDF at 35 mL/kg per hour versus 20 mL/kg per hour. At 90 days, mortality was 44.7% in both treatment groups (OR: 1.00; 95% CI: 0.81–1.23; $P = 0.99$). (From The RENAL Replacement Therapy Study Investigators. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361:1627–1638.)

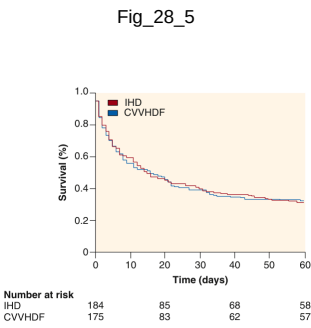


Fig. 28.8 60-Day Survival with Intermittent Hemodialysis (IHD) versus Continuous Venovenous Hemodiafiltration (CVVHDF) in the Hemofiltration Study. Kaplan-Meier plot of survival among 359 critically ill patients with acute kidney injury randomized to intermittent hemodialysis versus continuous venovenous hemodiafiltration. At 60 days survival was 31.5% among patients randomized to IHD versus 32.6% among patients randomized to CVVHDF ($P = 0.98$). (From Vonnahme C, Camus C, Combes A, et al. Continuous venovenous haemodiafiltration versus intermittent haemodialysis for acute renal failure in patients with multiple-organ dysfunction syndrome: a multicentre randomised trial. *Lancet*. 2006;368:379–385.)

Table 28.1 Causes of Prerenal Acute Kidney Injury	
Intravascular volume depletion	
Hemorrhage—trauma, surgery, postpartum, gastrointestinal	
Gastrointestinal losses—diarrhea, vomiting, nasogastric loss	
Renal losses—diuretics, osmotic diuresis, diabetes insipidus	
Skin and mucous membrane losses—burns, hyperthermia	
Nephrotic syndrome	
Cirrhosis	
Capillary leak	
Reduced cardiac output	
Cardiogenic shock	
Pericardial diseases—restrictive/constrictive/tamponade	
Congestive heart failure	
Valvular diseases	
Pulmonary diseases—pulmonary hypertension; pulmonary embolism	
Sepsis	
Systemic vasodilation	
Sepsis	
Cirrhosis	
Anaphylaxis	
Medications	
Renal vasoconstriction	
Early sepsis	
Hepatorenal syndrome	
Acute hypercalcemia	
Drugs—nonopiate, vasopressin, nonsteroidal, angiotensin-converting enzyme, calcineurin inhibitors, iodinated contrast agents	
Increased intraabdominal pressure	
Abdominal compartment syndrome	

Fig. 28_7 Table_28_1

Table 28.9 Major Causes of Acute Kidney Injury (AKI) in Specific Clinical Settings (Continued)
AKI and liver disease
Prerenal azotemia Reduced true (gastrointestinal hemorrhage, GI losses from lactulose, diuretics, large-volume paracentesis) circulatory volume or effective (hypoalbuminemia, splanchnic vasodilatation) Hepatorenal syndrome type 1 or 2 Tense ascites with abdominal compartment syndrome Intrinsic AKI Ischemic (severe hypoperfusion—see earlier) or direct nephrotoxicity and hepatotoxicity of drugs or toxins (e.g., carbon tetrachloride, acetaminophen, tetracyclines, and methoxyflurane) Tubulointerstitial nephritis + hepatitis caused by drugs (e.g., sulfonamides, rifampin, phenytoin, allopurinol, and phenindione), infections (leptospirosis, brucellosis, Epstein-Barr virus infection and cytomegalovirus infection), malignant infiltration (leukemia and lymphoma) or sarcoidosis Glomerulonephritis or vasculitis (e.g., polyarteritis nodosa, ANCA-associated GN, cryoglobulinemia, SLE, postinfectious hepatitis, or liver abscess)
AKI and nephrotic syndrome
Prerenal azotemia Intravascular volume depletion (diuretic therapy, hypoalbuminemia) Intrinsic AKI Manifestation of primary glomerular disease Collapsing glomerulopathy (e.g., HIV and pamidronate) Associated ATI (elderly hypertensive males) Associated interstitial nephritis (NSAIDs, rifampin, interferon alfa) Other—amyloid or light chain deposition disease, renal vein thrombosis, severe interstitial edema
ANCA, anti-neutrophil cytoplasmic antibody; ATI, acute tubular injury; GN, glomerulonephritis; HELLP, Hemolysis, Elevated Liver enzymes, Low Platelets; HIV, human immunodeficiency virus; HUS, hemolytic uremic syndrome; NSAID, nonsteroidal antiinflammatory drug; SLE, systemic lupus erythematosus.

Table_28_9_part2

Table 28.10 Cairo-Bishop Definition of Tumor Lysis Syndrome
Diagnosis of laboratory tumor lysis syndrome
Requires at least 2 of the following criteria achieved in the same 24-h interval from 3 days before to 7 days after chemotherapy initiation: • Uric acid: ≥8.0 mg/dL or ≥25% increase from baseline • Potassium: ≥6.0 mmol/L or ≥25% increase from baseline • Phosphorus: ≥4.6 mg/dL (≥6.5 mg/dL in children) or ≥25% increase from baseline • Calcium: ≤7.0 mg/dL or ≥25% decrease from baseline
Diagnosis of clinical tumor lysis syndrome
Laboratory tumor lysis syndrome plus at least 1 of the following: • Serum creatinine ≥1.5 times the age-adjusted upper limit of normal • Cardiac arrhythmia/sudden death • Seizure
Adapted from Cairo MS, Bishop M. Tumour lysis syndrome: new therapeutic strategies and classification. <i>Br J Haematol</i> . 2004;127:3–11.

Table_28_10

Table 28.11 Diagnostic Criteria for Hepatorenal Syndrome
• Cirrhosis with ascites • AKI defined as • ≥0.3 mg/dL increase in serum creatinine over <48 h • ≥50% increase in serum creatinine known or presumed to have occurred within prior 7 days • No improvement of serum creatinine (decrease to a level of ≤1.5 mg/dL) after at least 2 days of diuretic withdrawal and volume expansion with albumin (1 g/kg body weight per day to a maximum of 100 g/day) • Absence of shock • Absence of parenchymal kidney disease as indicated by • Proteinuria <500 mg/day • Microhematuria (>50 red blood cells per high-power field) and/or • Abnormal renal ultrasonography
Type 1 hepatorenal syndrome
Rapid progressive AKI defined based on increase in serum creatinine by >2× baseline
Type 2 hepatorenal syndrome
Moderate renal dysfunction with a steady or slowly progressive course
AKI, Acute kidney injury. Adapted from Angeli P, Gines P, Wong F, et al. Diagnosis and management of acute kidney injury in patients with cirrhosis: revised consensus recommendations of the International Club of Ascites. <i>Gut</i> . 2015;64:S31–S37.

Table_28_11

Table 28.12 Updated Classification of Hepatorenal Syndrome (HRS)		
Old terminology	New classification	Criteria
Type 1 HRS	HRS-AKI	• Absolute increase in serum creatinine of ≥ 0.3 mg/dL within 48 h and/or • Urine output ≤ 0.5 mL/kg body weight per hour for ≥ 6 h or • Relative increase in serum creatinine $\geq 50\%$ using the last available outpatient serum creatinine within 3 months as baseline
Type 2 HRS	HRS-AKD	• eGFR < 60 mL/min/1.73 m² for < 3 months in the absence of other causes • Relative increase in serum creatinine $< 50\%$ using the last available outpatient serum creatinine within 3 months as baseline
	HRS-CKD	• eGFR < 60 mL/min/1.73 m² for ≥ 3 months in the absence of other causes
AKD, acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate. Adapted from Angeli P, Garcia-Tsao G, Nadim MK, Parikh CR. News in pathophysiology, definition and classification of hepatorenal syndrome: a step beyond the International Club of Ascites (ICA) consensus document. <i>J Hepatol</i>. 2019;71:811–822.		

Table_28_12

Table 28.13 Complications of Acute Kidney Injury						
Metabolic	Cardiovascular	Gastrointestinal	Neurologic	Hematologic	Infectious	Other
Hyperkalemia	Pulmonary edema	Nausea	Neuromuscular irritability	Anemia	Pneumonia	Hiccups
Metabolic acidosis	Arrhythmias	Vomiting	Asterixis	Bleeding	Septicemia	Elevated parathyroid hormone
Hyponatremia	Pericarditis	Malnutrition	Seizures		Urinary tract infection	Low total triiodothyronine and thyroxine
Hypocalcemia	Pericardial effusion	Hemorrhage	Mental status changes			Normal thyroxine
Hyperphosphatemia	Pulmonary embolism					
Hypermagnesemia	Hypertension					
Hyperuricemia	Myocardial infarction					

Table_28_13

Table 28.14 Supportive Management Of Acute Kidney Injury	
Management issue	Treatment
Intravascular volume overload	Restriction of salt (<12 g/day) and water (<1 L/day) intake Diuretic therapy (if nonoliguric) Ultrafiltration
Hypotension	Restriction of oral and intravenous free water
Hypokalemia	Calcium gluconate (10 mL of 10% solution over 5 min if ECG changes are present) Glucose (5% mL of 50%) + insulin (0.1 U/kg) if needed Albumin (10–20 mg by reboiler or AKI)
Metabolic acidosis	Kidney replacement therapy Loop diuretics (if nonoliguric) IC binding resin Discontinue K ⁺ supplements or K ⁺ -sparing diuretics
	Restriction of dietary potassium Restriction of dietary protein Sodium bicarbonate (if HCO ₃ [−] <15 mmol/L)
Hypophosphatemia	Kidney replacement therapy Restriction of dietary phosphate intake Phosphate binding agents potassium hydroxide, calcium carbonate, calcium acetate, sevelamer, cerivite
Hypocalcemia	Oral or intravenous replacement 1,25-dihydroxyvitamin D ₃ or vitamin D ₃ (both known to be administered)
Hypermagnesemia	Discontinue magnesium-containing antacids
Malnutrition	Caloric intake: 20–35 kcal/day Protein intake Nonprotein requiring 0.8–1.0 g/kg/day Dietary protein: 1.0–1.5 g/kg/day
Drug dosages	Continuous kidney replacement therapy: up to 1.7 g/kg/day Intermittent hemodialysis: 1.0–1.5 g/kg/day Eradicate route of nutrition preferred Adjust all doses for CRRT and kidney replacement modality

ECG, Electrocardiogram; GFR, glomerular filtration rate; K⁺, potassium; K⁺ binders, potassium binders.

Table_28_14

Table 28.15 Effectiveness of Preventive Interventions for Contrast-Associated Acute Kidney Injury		
No benefit	Unclear benefit	Beneficial
Loop diuretics ^a	Statins	Isotonic intravenous fluids
Mannitol ^a	Natriuretic peptides	Low/iso-osmolal contrast media
Dopamine ^a	Theophylline/aminophylline	
Fenoldopam ^a	Ascorbic acid	
Hemodialysis ^a	Hemofiltration	
N-acetylcysteine		
Sodium bicarbonate ^b		
^a Potentially deleterious. ^b As compared with isotonic saline.		

Table_28_15

Table 28.16 Pharmacologic Interventions found to be Ineffective for Acute Tubular Injury in Clinical Trials
Diuretics Dopamine Fenoldopam Thyroid hormone Alpha melanocyte stimulating hormone Atrial natriuretic peptide Alkaline phosphatase Insulin growth factor Erythropoietin Prostaglandin A1

Table_28_16

Table 28.18 Clinical Trials: Timing of Initiation of Kidney Replacement Therapy (KRT) in Critically Ill Patients with Acute Kidney Injury (AKI)	ELAIN ¹⁰⁰	AKIK ¹⁰¹	IDEAL-ICU ¹⁰²	STARTRT-AKI ¹⁰³
Patients	231	620	488	2927
Centers	1	31	29	168
Population	Mixed ICU	Mixed ICU	Mixed ICU	Mixed ICU
Initial KRT modality	CVV-DIF	IHD (50%)/CRRT (45%)	IHD (54%)/CRRT (46%)	IHD (58%)/PIKRT (41%)/CRRT (8%)
AKI entry criteria	KIDGO stage 2	KIDGO stage 3	RIFLE-F	KIDGO stage 2 or 3
Additional inclusion criteria	Fluid overload Severe sepsis Vasopressor dependent Nonrenal organ dysfunction	Mechanical ventilation Vasopressor dependent	Septic shock	Age ≥18 Resolving cases in ICU Equipose among treating physicians
Criteria for delayed KRT	KIDGO stage 3	[K ⁺] >6.0 mmol/L Pulmonary edema pH <7.15 BUN >112 mg/dL Oligo-anuria >72 hours	[K ⁺] >6.5 mmol/L Pulmonary edema pH <7.20 or HCO ₃ ⁻ <12 mmol/L No recovery after 48 hours	[K ⁺] ≥6.0 mmol/L pH <7.20 or HCO ₃ ⁻ <12 mmol/L Respiratory failure No recovery after 72 hours
Time from enrollment to KRT (hours)	6.0	4.3	7.6	4.4
early	25.5	57	51.5	29.1
late	21	53	44	25
Patients in late group without KRT	9%	49%	38%	38.2%
Mortality (early vs. late)	30.4% vs. 40.3% (P = 0.11)	41.6% vs. 43.5% (P = 0.79)	45.1% vs. 42.1% (P = 0.48)	43.9% vs. 43.7% (P = 0.92)
Day 28	38.4% vs. 50.4% (P = 0.07)	48.8% vs. 49.7% (P = 0.79)	57.7% vs. 58.7% (P = 0.38)	
Day 90				
	38.4% vs. 50.4% (P = 0.03)			

AKIK: Artificial Kidney Initiation in Kidney Injury; CRRT: continuous kidney replacement therapy; ELAIN: Effect of Early vs. Delayed Initiation of Renal Replacement Therapy on Mortality in Critically Ill Patients with Acute Kidney Injury; ICU: intensive care unit; IDEAL-ICU: Initiation of Delayed Versus Early Initiation of Renal Replacement Therapy in Critically Ill Patients; IHD: intermittent hemodialysis; STARTRT-AKI: Standard versus Accelerated Initiation of Renal Replacement Therapy in Acute Kidney Injury.

Table_28_18